MTH 204 : Lecture 09

Case 2: Undamped Forced Oscillation $(c=0)$: Resonance

If $\omega = \omega_0$ (i.e. $\cos \omega t$ is a solution of the homogeneous equation), the situation is called resonance.

In this case, the particular solution found earlier is no longer valid.

Now, the ODE is $\quad my'' + ky = F_0 \cos(\omega_0 t)$

$$\Rightarrow y'' + \frac{k}{m} y = \frac{F_0}{m} \cos(\omega_0 t)$$

$$\Rightarrow y'' + \omega_0^2 y = \frac{F_0}{m} \cos(\omega_0 t)$$

Let $y_p(t) = t\left(A\cos(\omega_0 t) + B\sin(\omega_0 t)\right)$

$$y_p'(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t) - At\omega_0 \sin(\omega_0 t) + B\omega_0 t \cos(\omega_0 t)$$

$$= (A + B\omega_0 t)\cos(\omega_0 t) + (B - At\omega_0)\sin(\omega_0 t)$$

$$y_p''(t) = B\omega_0 \cos(\omega_0 t) - (A\omega_0 + B\omega_0^2 t)\sin(\omega_0 t)$$
$$- A\omega_0 \sin(\omega_0 t) + (B\omega_0 - A\omega_0^2 t)\cos(\omega_0 t)$$

$$= (2B\omega_0 - At\omega_0^2)\cos \omega_0 t - (Bt\omega_0^2 + 2A\omega_0)\sin(\omega_0 t)$$

Now if $y_p(t)$ satisfies the above equation

we get $(2B\omega_0 - At\omega_0^2)\cos(\omega_0 t) - (Bt\omega_0^2 + 2A\omega_0)\sin(\omega_0 t)$
$$+ \omega_0^2\left(tA\cos(\omega_0 t) + tB\sin(\omega_0 t)\right) = \frac{F_0}{m}\cos(\omega_0 t)$$

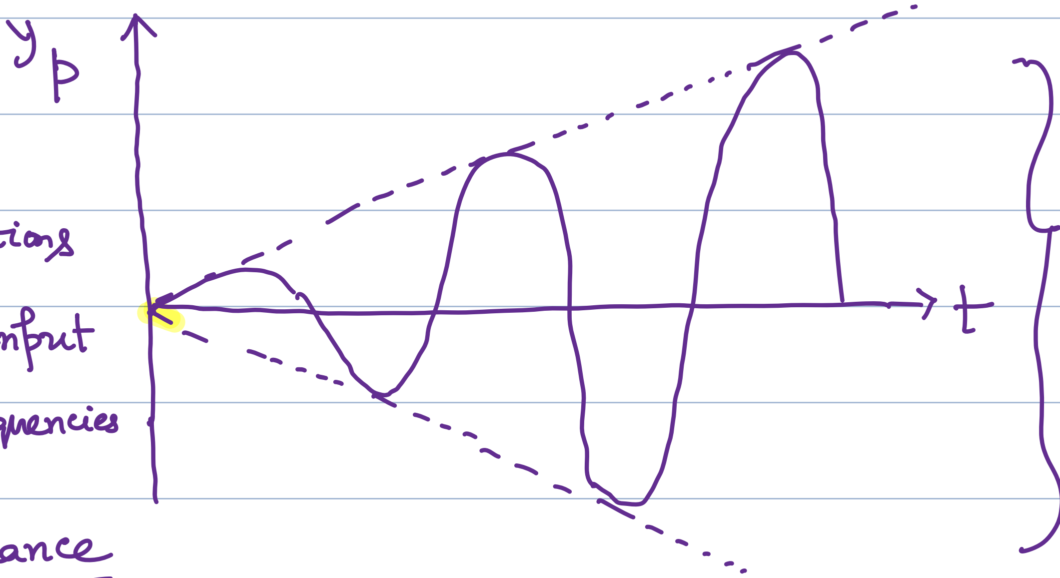
$$\Rightarrow 2B\omega_0 \cos(\omega_0 t) - 2A\omega_0 \sin(\omega_0 t) = \frac{F_0}{m}\cos(\omega_0 t)$$

$$\Rightarrow \quad 2B\omega_0 = \frac{F_0}{m} \quad \text{and} \quad A = 0$$

$$\Rightarrow \quad B = \frac{F_0}{2m\omega_0} \quad \text{and} \quad A = 0$$

So, $$\boxed{y_p(t) = \frac{F_0}{2m\omega_0}\, t \sin(\omega_0 t)}$$

This excitation of large oscillations by matching input and natural frequencies ($\omega = \omega_0$) is called <u>Resonance</u>



<u>Tacoma Bridge resonance</u>

Because of the factor t, the amplitude of the vibration becomes larger and larger.
Thus with very little damping, system may undergo large vibrations that can destroy the system.

<u>Case: Damped Forced Oscillations (Practical Resonance):</u>

In practice, there is always some damping and the amplitude of the oscillation doesn't grow infinitely.

Let us find the maximum amplitude in this case.

The particular solution of $my'' + cy' + ky = F_0 \cos(\omega t)$ was $y_p = a\cos(\omega t) + b\sin(\omega t)$

where $a = F_0 \dfrac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$ and $b = F_0 \dfrac{c\omega}{m^2(\omega_0^2-\omega^2)^2 + \omega^2 c^2}$

We can rewrite it as $y_p(t) = C_1 \cos(\omega t - \eta)$

where $C_1 = C_1(\omega) = \sqrt{a^2 + b^2} = F_0 \dfrac{1}{\sqrt{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}}$

and $\eta = \tan^{-1}\left(\dfrac{b}{a}\right)$

Now, for maximum amplitude $\dfrac{dC_1}{d\omega} = 0$

$$\Rightarrow F_0\left(-\frac{1}{2}\left(m^2(\omega_0^2-\omega^2)^2 + \omega^2 c\right)^{-\frac{3}{2}}\right)\left[2m^2(\omega_0^2-\omega^2)(-2\omega) + 2\omega c^2\right] = 0$$

$$\Rightarrow 2m^2(\omega_0^2 - \omega^2)(-2\omega) + 2\omega c^2 = 0$$

$$\Rightarrow c^2 = 2m^2(\omega_0^2 - \omega^2)$$

$$\Rightarrow 2m^2\omega^2 = 2m^2\omega_0^2 - c^2$$

$$\Rightarrow \omega^2 = \omega_0^2 - \frac{c^2}{2m^2}$$

$$\Rightarrow \boxed{\omega_{max}^2 = \omega_0^2 - \frac{c^2}{2m^2}}$$

It can be shown that $\left.\dfrac{d^2 C_1}{d\omega^2}\right|_{\omega_{max}}$

$= -\dfrac{1}{2}\dfrac{F_0 \quad 8m^2\omega^2}{\left[m^2(\omega_0^2-\omega^2)^2+\omega^2c^2\right]^{3/2}}$ Evaluated at $\omega = \omega_{max}$

< 0

That is Practical resonance occurs at a frequency slightly smaller than the natural frequency.

Now maximum amplitude at ω_{max} can be calculated as

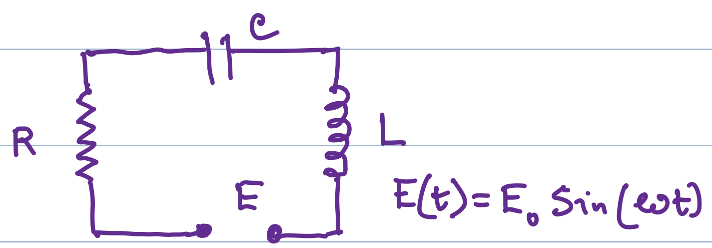
$$(c_1)_{max} = \frac{F_0}{\sqrt{m^2\left(\frac{c^2}{2m^2}\right)^2 + c^2\left(\omega_0^2 - \frac{c^2}{2m^2}\right)}} = F_0 \frac{2m}{c\sqrt{4m^2\omega_0^2 - c^2}}$$

— x — x — x — x — x — x —

<u>Note:</u> The general solution of NH: $y = y_h + y_p$ is called Transient solution. When all the roots of the characteristic equation of the homogeneous ODE are negative or have negative real parts, $y_h \to 0$ as $x \to \infty$ and the transient solution $y = y_h + y_p \to y_p$ (The steady state solution)

— x — x — x — x — x — x —

<u>Electric Circuit:</u>



Let $I(t)$ be the current in the circuit at time t and $Q(t)$ be the total charge in capacitor at time t.

Then $I(t) = \frac{dQ}{dt} \Rightarrow Q(t) = \int I(t)\,dt$.

The ODE modeling the RLC circuit is:

$$LI'(t) + RI(t) + \frac{1}{C}\int I(t)\,dt = E(t) = E_0 \sin(\omega t)$$

$$\Rightarrow LI'' + RI' + \frac{1}{C}I = E'(t) = E_0\omega\cos(\omega t)$$

$$\Rightarrow LI'' + RI' + \frac{1}{C}I = E_0\omega\cos(\omega t)$$

(H): $LI'' + RI' + \frac{1}{C}I = 0$

The characteristic equation is: $L\lambda^2 + R\lambda + \frac{1}{C} = 0$

$$\Rightarrow \lambda = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L} = -\frac{R}{2L} \pm \frac{1}{2L}\sqrt{R^2 - \frac{4L}{C}}$$

$$\Rightarrow I_H(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

(Now $I_H(t) \to 0$ as $t \to \infty$. In actual circuit after a relatively short time.)

(NH): Let $I_p(t) = A \cos \omega t + B \sin \omega t$

$$I_p'(t) = -A\omega \sin \omega t + B\omega \cos \omega t$$

and $I_p''(t) = -A\omega^2 \cos \omega t - B\omega^2 \sin \omega t$

Now if it is a solution of NH, then

$$L\left(-A\omega^2 \cos \omega t - B\omega^2 \sin \omega t\right) + R\left(-A\omega \sin \omega t + B\omega \cos \omega t\right) + \frac{1}{C}\left(A \cos \omega t + B \sin \omega t\right) = E_0 \omega \cos \omega t$$

$$\Rightarrow \left[\left(-L\omega^2 + \frac{1}{C}\right)A + R\omega B\right] \cos \omega t + \left[-R\omega A + \left(-L\omega^2 + \frac{1}{C}\right)B\right] \sin \omega t = E_0 \omega \cos \omega t$$

$$\Rightarrow \left(-L\omega^2 + \frac{1}{C}\right)A + R\omega B = E_0 \omega$$

and $-R\omega A + \left(-L\omega^2 + \frac{1}{C}\right)B = 0$

$$\Rightarrow \begin{pmatrix} -L\omega^2 + \frac{1}{C} & R\omega \\ -R\omega & -L\omega^2 + \frac{1}{C} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} E_0 \omega \\ 0 \end{pmatrix}$$

$$\Rightarrow \omega \begin{pmatrix} -L\omega + \frac{1}{C\omega} & R \\ -R & -L\omega + \frac{1}{C\omega} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \omega \begin{pmatrix} E_0 \\ 0 \end{pmatrix}$$

$$\Rightarrow A = \frac{-E_0 S}{R^2+S^2}, \quad B = \frac{E_0 R}{R^2+S^2}$$

where S is the reactance, $S = L\omega - \frac{1}{C\omega}$

(The quantity $\sqrt{R^2+S^2}$ is called impedance or apparent resistance)

Thus $I_p(t) = (A\cos \omega t + B\sin \omega t)$

$$= \sqrt{A^2+B^2}\,\sin\left(\omega t + \tan^{-1}\frac{A}{B}\right)$$

$$\Rightarrow \boxed{I_p(t) = \frac{E_0}{\sqrt{R^2+S^2}} \sin\left(\omega t - \tan^{-1}\frac{S}{R}\right)}$$

Ex: Find the current $I(t)$ in an RLC circuit with $R = 3.6\,\Omega$, $L = 0.2\,H$, $C = 0.0625\,F$ which is connected to a source of $E(t)$

$E(t) = 164\cos 10t\ V$

Assume that the current and capacitor charge are zero when $t=0$ i.e. $I(0)=0$ and $Q(0)=0$

The differential equation is $LI'' + RI' + \frac{1}{C}I = -1640\sin 10t$

$$\Rightarrow .2I'' + 3.6I' + \frac{1}{.0625}I = -1640\sin 10t$$

$\Rightarrow .2I'' + 3.6I' + 16I = -1640\sin 10t$

The Homogeneous Equation:

$$.2I'' + 3.6I' + 16I = 0$$

$$\Rightarrow 2I'' + 36I' + 160I = 0$$

$$\Rightarrow I'' + 18I + 80 = 0$$

Characteristic Equation: $\lambda^2 + 18\lambda + 80 = 0$

$$\Rightarrow (\lambda+8)(\lambda+10) = 0$$

$\Rightarrow \lambda = -8, -10$ So, $y_h = c_1 e^{-8t} + c_2 e^{-10t}$

Now let $I_p = A\cos 10t + B\sin 10t$, Then $I_p' = -10A\sin 10t + 10B\cos 10t$

$$I_p'' = -100A\cos 10t - 100B\sin 10t$$

Hence $.2(-100A\cos 10t - 100B\sin 10t) + 3.6(-10A\sin 10t + 10B\cos 10t)$

$$+16(A\cos 10t + B\sin 10t) = -1640\sin 10t$$

$\Rightarrow (-20A + 36B + 16A)\cos 10t + (-20B - 36A + 16B)\sin 10t = -1640\sin 10t$

$\Rightarrow$ $-4A + 36B = 0 \Rightarrow A = 9B$

and

$-36A - 4B = -1640 \Rightarrow -36(9B) - 4B = -1640$

$$\Rightarrow -328B = -1640 \Rightarrow \boxed{B = 5}$$

Then $A = 9B = 5\times 9 = \boxed{45}$

So, $I(t) = c_1 e^{-8t} + c_2 e^{-10t} + 45\cos 10t + 5\sin 10t$

Now $I(0) = 0 \Rightarrow c_1 + c_2 + 45 = 0$

$$\Rightarrow c_1 + c_2 = -45 \quad \ldots\ldots\ldots (1)$$

Also $Q(0) = 0$

Now $LI'(t) + RI(t) + \frac{1}{C}Q(t) = 164\cos 10t$

When $t=0$, $\quad LI'(0) + RI(0) + \frac{1}{C}Q(0) = 164$

Then $\quad LI'(0) + 0 + 0 = 164$

$$\Rightarrow I'(0) = \frac{164}{L} = \frac{164}{.2} = 820$$

Now, $\quad I'(t) = -8c_1 e^{-8t} - 10c_2 e^{-10t} - 450\sin 10t + 50\cos 10t$

$$\Rightarrow I'(0) = -8c_1 - 10c_2 + 50$$

Therefore $\quad -8c_1 - 10c_2 + 50 = 820$

$$\Rightarrow -8c_1 - 10c_2 = 770 \quad \ldots\ldots (2)$$

Now (1) x 8 $\Rightarrow$ $\quad 8c_1 + 8c_2 = -360$

(2) x 1 $\Rightarrow$ $\quad -8c_1 - 10c_2 = 770$

Adding $\quad -2c_2 = 410$

$$\Rightarrow \boxed{c_2 = -205}$$

Then (1) $\Rightarrow c_1 + c_2 = -45 \Rightarrow c_1 - 205 = -45$

$$\Rightarrow \boxed{c_1 = 160}$$

$$\boxed{\text{Hence } I(t) = 160e^{-8t} - 205e^{-10t} + 45\cos 10t + 5\sin 10t}$$

Method of Variation of Parameter:

(Lagrange)

$$y'' + p(x) y' + q(x) y = r(x)$$

p, q, r don't need to be constants but they are continuous function on an open interval I.

Let y_1 and y_2 be two independent solution of the associated homogeneous equation.

Let us assume that there is a particular solution of the NH equation of the form

$$y_p = u(x) y_1 + v(x) y_2$$

$$y_p' = u' y_1 + u y_1' + v' y_2 + v y_2'$$
$$= (u' y_1 + v' y_2) + (u y_1' + v y_2')$$

Let us impose the condition that $\boxed{u' y_1 + v' y_2 = 0}$

Then $y_p' = u y_1' + v y_2'$

$$\Rightarrow y_p'' = u y_1'' + u' y_1' + v y_2'' + v' y_2'$$
$$\Rightarrow y_p'' = u y_1'' + v y_2'' + u' y_1' + v' y_2'$$

Now $y_p'' + p(x) y_p' + q(x) y_p = r(x)$

$$\Rightarrow \left(u y_1'' + v y_2'' + u' y_1' + v' y_2'\right) + \left(u y_1' + v y_2'\right) p(x)$$
$$+ \left(u y_1 + v y_2\right) q(x) = r(x)$$

$$\Rightarrow \left[y_1'' + p(x) y_1' + q(x) y_1\right] u(x)$$
$$+ \left[y_2'' + p(x) y_2' + q(x) y_2\right] v(x)$$
$$+ u' y_1' + v' y_2' = r(x)$$

$$\Rightarrow \boxed{u' y_1' + v' y_2' = r(x)}$$

So, $$\boxed{\begin{aligned} u' y_1 + v' y_2 &= 0 \\ u' y_1' + v' y_2' &= r(x) \end{aligned}}$$

$$\Rightarrow \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} 0 \\ r(x) \end{pmatrix}$$

$$\Rightarrow u' = \frac{-r y_2}{W}, \quad v' = \frac{r y_1}{W}$$

where $W = y_1 y_2' - y_2 y_1'$
(Wronskian)

$$\Rightarrow \quad u = -\int \frac{r y_2}{W}\, dx$$

$$v = \int \frac{r y_1}{W}\, dx$$

So, $\boxed{y_p = -y_1 \int \frac{r y_2}{W}\, dx + y_2 \int \frac{r y_1}{W}\, dx}$

$$\boxed{\text{Hence } y = C_1 y_1 + C_2 y_2 + (-y_1) \int \frac{r y_2}{W}\, dx + y_2 \int \frac{r y_1}{W}\, dx}$$

Ex: $\quad y'' + y = \csc x$

Note that

$y_1 = \cos x$ and $y_2 = \sin x$ are independent

Solutions of $\quad y'' + y = 0$

(characteristic equation: $\lambda^2 + 1 = 0 \Rightarrow \lambda = \pm i$)

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$y_p = -y_1 \int \frac{r y_2}{W} dx + y_2 \int \frac{r y_1}{W} dx$$

$$= -\cos x \int \frac{1}{\sin x} \cdot \frac{\sin x}{1} dx + \sin x \int \frac{1}{\sin x} \cdot \frac{\cos x}{1} dx$$

$$= -x \cos x + \sin x \ln|\sin x|$$

So, the general solution is:

$$y = y_h + y_p = c_1 \cos x + c_2 \sin x + \sin x \ln|\sin x| - x \cos x$$

$$= \boxed{(c_1 - x)\cos x + \left(c_2 + \ln|\sin x|\right)\sin x}$$

The solution is valid on the intervals where $\csc x$ and $\ln|\sin x|$ are defined (i.e. when $\sin x \neq 0$)

Ex: $y'' - 2y' + y = e^x \sin x$

Homogeneous equation: $y'' - 2y' + y = 0$

characteristic equation: $\lambda^2 - 2\lambda + 1 = 0 \Rightarrow (\lambda - 1)^2 = 0$

Thus there is a real double root $\lambda = 1$

So, two independent solutions of the homogeneous equation are $y_1 = e^x$, $y_2 = x e^x$

$$W = \begin{vmatrix} e^x & xe^x \\ e^x & xe^x + e^x \end{vmatrix} = e^{2x} + xe^{2x} - xe^{2x} = e^{2x}$$

So, $$y_p = -e^x \int \frac{e^x \sin x \, xe^x}{e^{2x}} dx + xe^x \int \frac{e^x \sin x \, e^x}{e^{2x}} dx$$

$$= -e^x \int x \sin x \, dx + xe^x \int \sin x \, dx$$

$$= -e^x \left[-x\cos x + \int \cos x \, dx \right] + xe^x(-\cos x)$$

$$= \cancel{xe^x \cos x} - e^x \sin x - \cancel{xe^x \cos x} = -e^x \sin x$$

So, the general solution is

$$\boxed{y = y_h + y_p = (c_1 + c_2 x)e^x - e^x \sin x}$$

Note: Can also be done by method of undetermined coefficient.